

Hyperactive Minority Alter the Stability of Community Notes

Keywords: Community-based fact-checking, misinformation, social media, polarization, computational social science

1 Extended Abstract

As platforms scale down professional fact-checking, community-based alternatives like Community Notes (CN) on X are promoted as more transparent and scalable [1, 9]. While research shows crowd-sourced approaches can achieve accuracy comparable to experts [7, 3], concerns emerge about timing sensitivity [8], limited coverage [5], and partisan asymmetries [6]. Yet how social dynamics—activity concentration and polarization typical of social media [2, 4]—shape note emergence remains understudied.

We analyze the complete public dataset (January 2021-December 2025). Crucially, we integrate the open-source CN consensus algorithm, enabling counterfactual simulations assessing how rater composition affects note visibility.

RQ1: Activity Concentration. Figure 1 reveals extreme concentration following a power-law ($\alpha = 3.06$): 20% of contributors produce 80% of ratings (Gini $G = 0.78$). Moreover, raters systematically focus on limited author sets, displaying strong selectivity in content evaluation.

RQ2: Political Polarization. We construct a behavioral proxy for political leaning, defining $L \in [-1, 1]$ where positive values indicate Republican-leaning and negative Democratic-leaning behavior. While aggregate distribution is nearly symmetric ($\mu_L = -0.08$), Figure 2 reveals striking differences by activity level. The top activity decile exhibits substantially more polarized behavior with clear Republican shift. Ashman’s D (bimodality measure) increases with activity confirming the most active users—those with greatest influence—are disproportionately polarized.

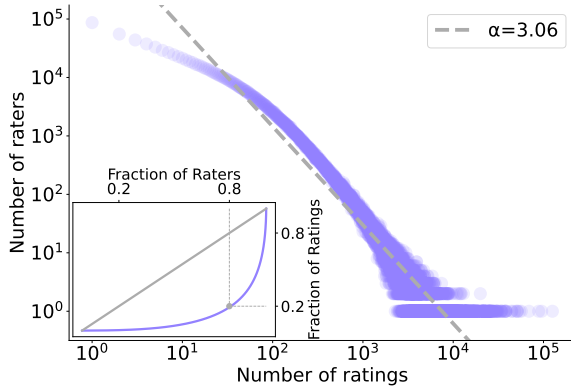


Figure 1: Log-log distribution of ratings per contributor with Lorenz curve showing 20% of users account for 80% of ratings.

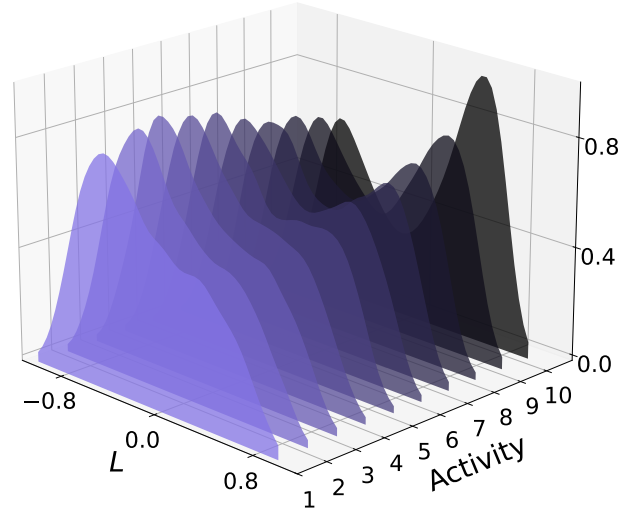


Figure 2: Political leaning distribution across activity deciles. Polarization increases markedly with user activity.

RQ3: System Instability. We construct counterfactuals removing top 10-10,000 most active raters (only $7.69 \times 10^{-6}\%$ to $7.69 \times 10^{-3}\%$ of contributors) and measure stability via Jaccard similarity. Figure 3 reveals substantial fragility: removing merely 10 users causes 4,587 of 152,565 Currently Rated Helpful notes—those visible to all X users—to lose visibility. Removing 10,000 drops Jaccard similarity to 69%. A tiny group of hyperactive, polarized contributors exerts disproportionate influence over which notes emerge.

Conclusion. Community Notes replicates social media dynamics: extreme activity concentration and political polarization among the most active. These participation asymmetries produce structural instability—despite the algorithm requiring cross-partisan agreement, a small polarized minority disproportionately controls which notes become visible. Rather than decentralizing epistemic authority, CN reconfigures it. While praised for transparency,

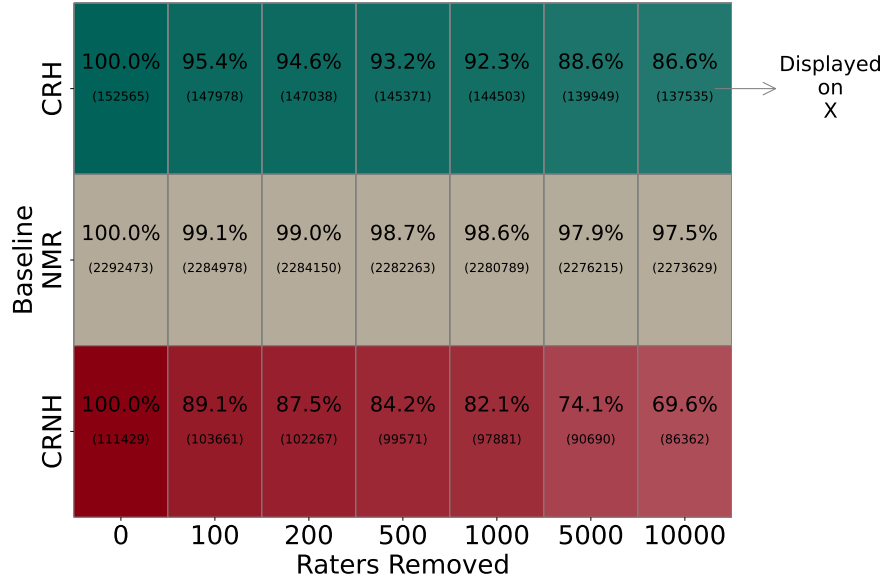


Figure 3: Note status under counterfactuals. Removing tiny fractions of hyperactive users substantially alters visible notes.

realizing the "wisdom of the crowd" proves difficult when participation is highly unequal and politically asymmetric. Future work should explore algorithmic modifications for robustness, such as capping individual activity or down-weighting hyperactive contributors.

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